3.6 Graphing Linear Equations in Slope-Intercept Form

Name _____ Period ____ Date _____

Algebra I 3.6 · Student sheet

I can…

- find the slope of a line from a graph, two points, or a table.
- identify the slope and y-intercept and graph y = mx + b.
- interpret the slope and y-intercept in real-life situations.

Vocabulary · fill in the missing word

Slope — m = _____/run = $\frac{y_2 - y_1}{x_2 - x_1}$

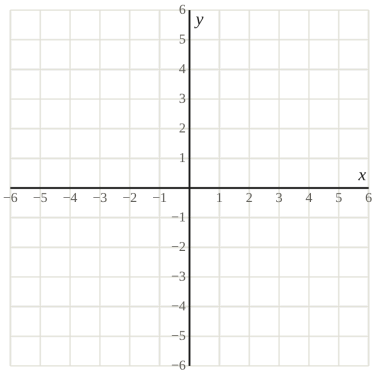
Slope-intercept form — y = mx + b: slope m, _____ b

Kinds of slope — _____ (rises), negative (falls), zero (horizontal), undefined (vertical)

Part 1 · Walk the line

Graph all three lines: start at the y-intercept, then use the slope to walk to more points.

y = 2x + 1 y = −x + 4 y = ½x − 2



Line	Slope	Positive or negative?
y = 2x + 1		
y = −x + 4		
y = ½x − 2		

Part 2 · Notes and examples (fill in each note, then work the example)

Slope: Slope m = _____ / _____ = $(y_2 - y_1)/($_____$)$

Special slopes: A horizontal line has slope _____; a vertical line has _____ slope.

Example 1

Find the slope of the line through (−2, 5) and (4, −1).

Slope-intercept form: In y = mx + b, m is the _____ and b is the _____.

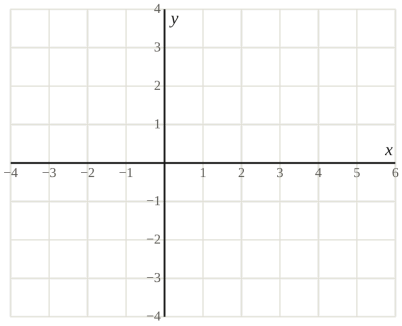
Example 2

Find the slope and y-intercept.

a. $y = -3x + 4$ b. $2x + y = 5$

Example 3

Graph $y = (\frac{2}{3})x - 1$. Find the x-intercept.



Example 4

The cost of a party is $y = 12x + 50$, where x is the number of guests. What do the slope and y-intercept mean?

Part 3 · Partner practice

Partner 1

Find the slope from the table: x: 0, 1, 2; y: 7, 4, 1

Partner 2

Find the slope and y-intercept of $y = 6 - x$.